Fig. 20: The Accelerator-6D PCIe form-factor accelerator board (Source: Achronix)

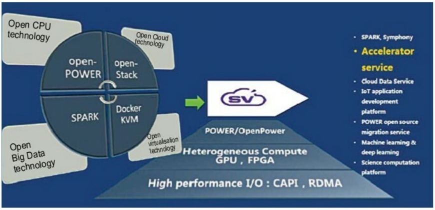

Fig. 21: SuperVessel platform

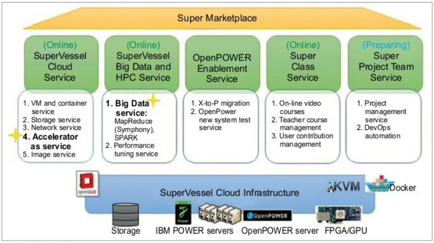

Fig. 22: SuperVessel roadmap

die. It aims to put both the chips on a single die as early as next year. Both CPUs and FPGAs require a lot of power. Putting the two in one socket will limit performance to the thermal envelope of the single chip.

Microsoft is using Altera FPGAs both in its networking cards and as accelerators for its Bing search. China server vendor Inspur is trying to ride the GPU train with a server packing 16 graphics processors. The server uses a message passing interface the company created for the Caffe deep learning framework developed at U.C. Berkeley for use with GPU accelerators. FPGAs accelerate the ranking of search terms—a new partition of the overall search job.

A special embedded core makes the FPGA more programmable. China's Baidu has developed an FPGA board accelerating at least ten low-level tasks. The PCIe 2.0 card uses a Xilinx K7 480t with 4GB memory.

Speedster 22i FPGAs built on Intel's advanced 22nm process technology implement all of interface functions as hardened IP. This results in lower power consumption, reduced programmable logic fabric, faster clock rates, and simpler design since the interface IP is already timing-closed.

Achronix Accelerator-6D board is said to offer the highest memory bandwidth for an FPGA-based PCIe form factor board. This PCIe add-in card is well suited for high-speed data centre acceleration applications. The Accelerator-6D packs a Speedster22i HD1000 FPGA with 700,000 lookup tables, which connects to six independent memory controllers to allow up to 192GB of memory and 690Gbps of total memory bandwidth. Accelerator-6D board comes with a power supply, one-year licence for Achronix ACE design tools, and multiple system-level reference designs for Ethernet, DDR3 and PCIe operation.

Companies are collaborating on the specification for the new Cache Coherent Interconnect for Accelerators (CCIX). A single interconnect technology specification will ensure that processors using different instruction-set architectures (ISAs) can coherently share data with accelerators and enable efficient heterogeneous computing, significantly improving compute efficiency for servers running data centre workloads. CCIX will allow these components to access and process data irrespective of where it resides, without the need for complex programming environments. This will enable both off-load and bump-in-the-wire inline application acceleration while building on existing server ecosystems and form